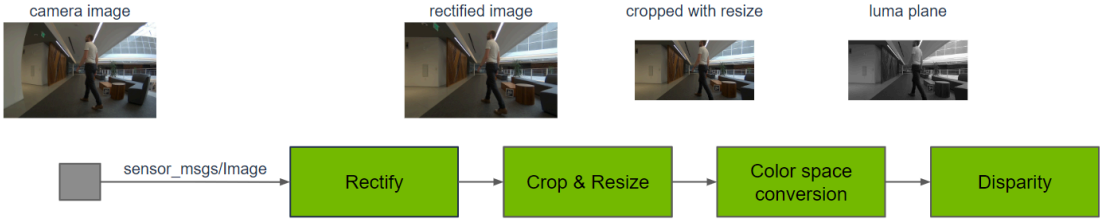
Isaac ROS Image Distortion Processing

Isaac ROS Image Distortion Processing
- Overview
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Isaac ROS Image Distortion Processing official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_image_pipeline/index.html

Overview

Isaac ROS Image Pipeline provides image preprocessing functions, including cropping, resizing, lens distortion correction, flipping, and color space conversion. A stereo camera can also combine the left and right images to generate disparity and point clouds.



Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Each terminal should use the same ROS/FastDDS environment. In the processing chain, `/image` and `/camera_info` are input to `RectifyNode`, which outputs `/image_rect` and `/camera_info_rect`, and `ResizeNode` then outputs `/resize/image`.

Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

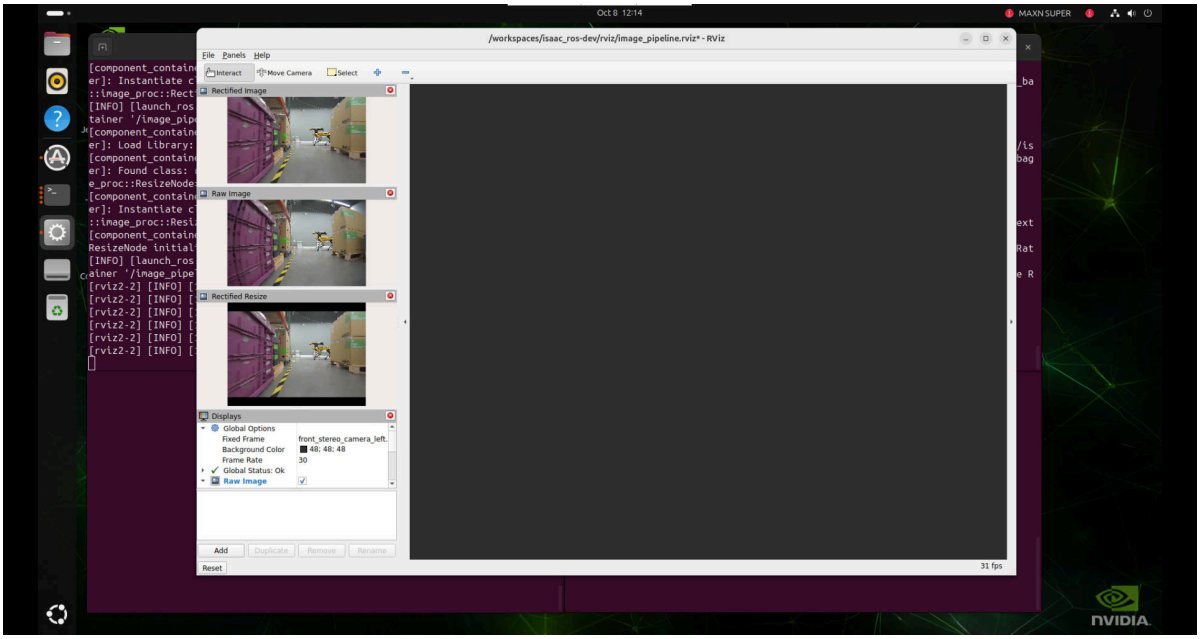
Terminal A: Start the processing chain and RViz:

```
ros2 launch /workspaces/isaac_ros-dev/launch/image_pipeline.launch.py
```

The RViz configuration is `/workspaces/isaac_ros-dev/rviz/image_pipeline.rviz`, which displays `/image`, `/image_rect`, and `/resize/image`.

Terminal B: Play the input bag in a loop:

```
ros2 bag play --loop --rate 0.5 \
  /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_visual_slam/quickstart_bag
\
  --remap \
  /front_stereo_camera/left/image_raw:=/image \
  /front_stereo_camera/left/camera_info:=/camera_info
```



Terminal C: Verify the input, undistorted, and resized outputs:

```
ros2 topic echo --once /image --field width
ros2 topic echo --once /image_rect --field width
ros2 topic echo --once /resize/image --field width
ros2 topic echo --once /camera_info_rect --field d
```